SINGLE OBSTRUCTION THEORY: PROJECTION NONLINEARITY, ℓ^1 RIGIDITY, AND DENSITY SCALING

JEREMY H. CARROLL

Abstract. Projection-based approximations—such as mean-field theory, product-state ansätze, and factorized variational methods—replace correlated states with products of local states via a nonlinear projection. We study the structural obstruction introduced by this projection.

We prove three results. **(1)** A nonlinear projection composed with a linear semigroup cannot be conjugate to any linear semigroup (Theorem 3.1). **(2)** The resulting defect admits a unique local additive diagnostic: the ℓ^1 coboundary norm, characterized as the free additive faithful contractive seminorm on the space of edge defects (Theorem 4.3). **(3)** For finite-range non-commuting interactions on finite-dimensional lattice systems, the per-bond obstruction density remains bounded below over a finite time interval whose duration is $O(1)$ in system size (Theorem 5.2).

Together these theorems establish **existence, uniqueness, and finite-time persistence** of projection obstructions within a presheaf-theoretic framework over finite posets with Banach-space stalks.

1. Introduction

Many-body computation relies on factorization: replace an exponentially large state with a product of local states. Mean-field theory, Hartree–Fock, product-state ansätze, and factorized variational methods all implement this via a **projection** Π onto the manifold of product states. This paper asks: *What does the projection discard, how should we measure it, and how does the discarded quantity scale?*

Main Result (informal). Projection-based factorization introduces a geometric curvature term in the induced dynamics on the product-state manifold. This curvature prevents linear conjugacy, manifests as a presheaf coboundary defect, and is measured uniquely by the ℓ^1 norm on edge defects. For finite-range lattice systems this obstruction appears with non-vanishing density over finite time intervals.

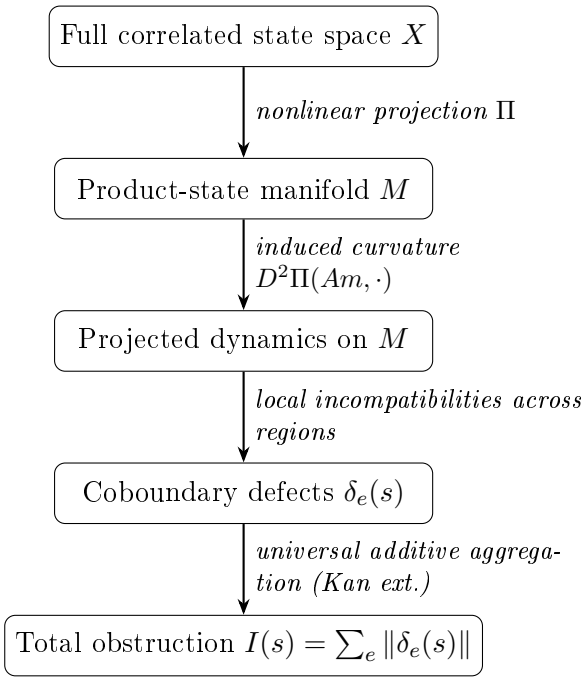
We prove three structural theorems governing this loss of information:

Date: March 2026.
Version 1.0 (preprint).

1. **Geometric Obstruction (Existence):** Π introduces affine curvature that prevents the projected dynamics from being conjugate to any linear semigroup (Theorem 3.1).
2. **Categorical Rigidity (Uniqueness):** The ℓ^1 coboundary norm is the universal local faithful diagnostic for this obstruction, arising as the free additive seminorm on the space of local defects (Theorem 4.3).
3. **Density Scaling (Persistence):** The per-bond obstruction density is $\Omega(1)$ in system size over an $O(1)$ time window, protected by Lieb–Robinson bounds (Theorem 5.2).

Scope. The framework applies rigorously to Banach presheaves over finite posets. We avoid speculative physics or infinite-time limits; the density scaling bound is finite-time only, as thermalization destroys local correlations at late times. The focus of this paper is to stabilize the core mathematical invariants of projection obstructions.

1.1. **Conceptual Overview.** The structure of the argument is summarized in the following diagram:



Interpretation:

- **Theorem 3.1:** Nonlinear projection introduces curvature that prevents linear conjugacy of the projected dynamics.
- **Theorem 4.3:** The resulting incompatibilities form local coboundary defects whose unique additive diagnostic is the ℓ^1 norm.
- **Theorem 5.2:** In lattice systems with finite-range interactions, these defects appear with nonzero density over finite time intervals.

2. Structural Framework

We work with Banach-valued presheaves over finite posets, following standard categorical conventions [3, 4].

2.1. **Presheaves over Finite Posets.** Let $(P, \leq)$ be a finite poset with covering relations $\mathrm{E}_{\mathrm{cov}}(P) = \{(a, b) : a < b,\ \nexists c \text{ with } a < c < b\}$. A **Banach presheaf** F over P assigns to each $x \in P$ a Banach space $F(x)$ and to each $(a, b) \in \mathrm{E}_{\mathrm{cov}}$ a contractive linear restriction map $r_{b,a}\colon F(b) \to F(a)$ with $\|r_{b,a}\|_{\mathrm{op}} \leq 1$. Restriction maps along general order relations $a < c$ are obtained by composition of covering restrictions.

2.2. **The Coboundary.** For each edge $e = (a, b)$, the **coboundary** is $\delta_e(s) = r_{b,a}(s(b)) - s(a) \in F(a)$. A section descends if $\delta_e(s) = 0$ for all e. The ℓ^1 coboundary norm is $I(s) = \sum_{e \in \mathrm{E}_{\mathrm{cov}}} \|\delta_e(s)\|$.

2.3. **Categorical Interpretation via Kan Extensions.** The obstruction functional has a canonical origin in category theory. Interpret the covering edges as a category $\mathrm{Edge}(P)$ with a projection functor $\pi\colon \mathrm{Edge}(P) \to P$ sending each edge $e = (a < b)$ to its lower vertex a. For a presheaf F, the local defect assignment $d(s)(e) = \delta_e(s)$ defines a section of the pulled-back presheaf $F \circ \pi^{\mathrm{op}}$.

The natural way to aggregate these dispersed local defects globally is the left Kan extension along π. Because the ℓ^1 direct sum realizes the categorical coproduct in the category of Banach spaces with contractive maps, the colimit evaluates to the space of edge defects $\mathrm{Lan}_\pi(F \circ \pi^{\mathrm{op}}) \cong \bigoplus_{e \in \mathrm{E}_{\mathrm{cov}}} F(a_e)$ equipped with the ℓ^1 norm. The total obstruction $I(s) = \|\mathrm{Lan}_\pi(d(s))\|_1$ is exactly the universal additive seminorm evaluated on this Kan extension.

2.4. **Linear Semigroups and Nonlinear Projections.** A C_0 **contraction semigroup** on a Banach space X is $\{T_t\}_{t \geq 0}$ with $T_0 = \mathrm{id}$, $T_{t+s} = T_t T_s$, $\|T_t\| \leq 1$, and strong continuity. Its generator is $(A, \mathcal{D}(A))$. A **nonlinear projection** is a continuous idempotent $\Pi\colon X \to M \subset X$ with M a regular submanifold and Π not affine.

3. Theorem I: Projection Non-Dilability

The first result identifies a geometric obstruction introduced by nonlinear projection.

3.1. **Statement.**

Theorem 3.1 (Projection Non-Dilability)**.** *Let X be a Banach space, $\{T_t\}$ a C_0 contraction semigroup with generator $(A, \mathcal{D}(A))$, and $\Pi\colon X \to M \subset X$ a C^2 idempotent projection onto a nonlinear submanifold $M \subset \mathcal{D}(A)$. Suppose:*

(a) T_t does not preserve M for some $t > 0$, and

(b) (Non-degeneracy) *There exists* $m_0 \in M$ *such that* $D^2\Pi(m_0)(Am_0, \cdot) \neq 0$.

Then there is no C^2 *conjugacy connecting* $T_t|_M$ *to any linear contraction semigroup. That is, there does not exist a Banach space* Y*, a* C^2 *embedding* $\iota\colon M \hookrightarrow Y$*, and a* C_0 *contraction semigroup* $\{\tilde{T}_t\}$ *on* Y *such that* $\Pi \circ T_t|_M = \iota^{-1} \circ \tilde{T}_t \circ \iota$.

3.2. **Preliminary Lemmas.**

Lemma 3.2 (Parallel fields on flat connections)**.** *Let* (∇, M) *be a connected Banach manifold equipped with a flat affine connection* ($R^\nabla = 0$)*. If a* $(1,1)$*-tensor field* S *on* M *satisfies* $\nabla S = 0$ *(parallel), then* S *is constant in any affine coordinate chart.*

Proof. Since $R^\nabla = 0$, the exponential map at any $m \in M$ is a local diffeomorphism onto a neighborhood in which the Christoffel symbols vanish [12, Prop. VII.5.2]. In these coordinates, $\nabla S = 0$ reduces to $\partial_i S^j_k = 0$, so the components are constant. On a connected manifold, two affine coordinate charts overlap via affine transitions (compositions of linear maps and translations), which preserve constancy. Therefore S is constant on all of M. □

Lemma 3.3 (Genericity of non-degeneracy)**.** *Let* M *be a finite-dimensional* C^2 *submanifold of a Banach space* X*,* A *a bounded linear operator on* X*, and* $\Pi\colon X \to M$ *a* C^2 *idempotent. Define* $\Phi\colon M \to \mathcal{L}(X; T_mM)$ *by* $\Phi(m) = D^2\Pi(m)(Am, \cdot)$*. If* Π *is not affine (i.e.,* $D^2\Pi \not\equiv 0$*), then the zero set* $Z = \Phi^{-1}(0)$ *has empty interior in* M*.*

Proof. Φ is continuous (Π is C^2, A is bounded). If Z had nonempty interior, Φ would vanish on an open set $U \subset M$. On U, $D^2\Pi(m)(Am, v) = 0$ for all $v \in T_mM$ and all $m \in U$. This implies $D^2\Pi|_U \equiv 0$, meaning Π is exactly affine when restricted to U. By unique continuation of C^2 maps on connected manifolds, Π would be affine on all of M, thereby contradicting the strict non-affinity hypothesis. □

3.3. **Proof of Theorem 3.1. Step 1: Linear flows are affinely flat.** A linear semigroup $\{\tilde{T}_t\}$ on Y with generator $\tilde{A}$ satisfies $\tilde{A}(y) = \tilde{A}y$. The canonical flat connection $\tilde{\nabla}$ on Y (given by $\tilde{\nabla}_v w = Dw \cdot v$) has $R^{\tilde{\nabla}} = 0$, and $\tilde{\nabla}^2 \tilde{A} = 0$.

Step 2: Conjugacy transfers flatness. If a C^2 embedding $\iota\colon M \hookrightarrow Y$ conjugates S_t to $\tilde{T}_t$, then $\nabla = \iota^*\tilde{\nabla}$ is a flat connection on M ($R^\nabla = 0$, since pullback preserves flatness [12, §VII.3]), and $\nabla^2 B = 0$ for the generator $B = \iota_*^{-1}\tilde{A}$ of S_t.

Step 3: The projected generator has non-constant covariant derivative. The projected generator is $B(m) = D\Pi(m) \cdot Am$ (well-defined since Π is idempotent, so $D\Pi(m)$ projects onto T_mM). Differentiating:

$$\nabla_v B = D^2\Pi(m)(Am, v) + D\Pi(m) \cdot Av.$$

If $\nabla^2 B = 0$, then $\nabla_v B$ is a parallel tensor field. By Lemma 3.2 (applicable since ∇ is flat), $\nabla_v B$ is constant in affine coordinates. But $D^2\Pi(m_0)(Am_0, \cdot) \neq 0$ (hypothesis (b)), and $m \mapsto D^2\Pi(m)(Am, \cdot)$ is continuous and nonzero on an open neighborhood U of m_0 (Lemma 3.3 with A bounded). Since the projection curvature $D^2\Pi(m)(Am, v)$ varies nontrivially with m on U, the tensor $\nabla_v B$ cannot be constant. Contradiction. □

Remark 3.4 (Second fundamental form). The curvature term $\mathrm{II}(Am, v) = D^2\Pi(m)(Am, v)$ is the second fundamental form of M evaluated along the generator. Theorem 3.1 restates as: *nonzero second fundamental form along the generator obstructs linear conjugacy.*

Remark 3.5 (Feedback linearization). The obstruction is structurally analogous to Jakubczyk–Respondek conditions [11] in nonlinear control: both involve second-derivative curvature terms preventing coordinate-based linearization. Here, the original dynamics are linear—projection introduces the nonlinearity.

4. Theorem II: Categorical Rigidity

4.1. **Diagnostic Systems.**

Definition 4.1. A **diagnostic system** is a family $\{J_P\}_{P \in \mathbf{Poset}_{\mathrm{fin}}}$ with $J_P \colon \Gamma(F_P) \to \mathbb{R}_{\geq 0}$ for each finite poset P and Banach presheaf F_P.

4.2. **Axioms.**

Axiom 1 (Coboundary Locality)**.** $J_P(s) = \sum_e j_e(\delta_e s)$, with $j_e \colon F(a) \to \mathbb{R}_{\geq 0}$.

Axiom 2 (Faithfulness)**.** $j_e(\delta) = 0 \iff \delta = 0$.

Axiom 3 (Contractive Monotonicity)**.** For any contraction $\varphi \colon V \to W$, $j_e(\varphi(\delta)) \leq \|\varphi\|_{\mathrm{op}} \cdot j_e(\delta)$.

Axiom 4 (Isomorphism Invariance)**.** For any poset isomorphism π, $J_{\tilde{P}}(\pi^* s) = J_P(s)$.

Remark 4.2 (Continuity). Axiom 3 implies j_e is Lipschitz: apply it to the Hahn–Banach functional f_δ and the embedding ι_δ (both contractions) to obtain $j_e(\delta) = j_e(\|\delta\|)$, and to scaling maps to obtain $j_e(c\delta) = c \cdot j_e(\delta)$. No additional regularity hypothesis is needed.

4.3. **Categorical Rigidity.**

Let **Ban** be the symmetric monoidal category of Banach spaces equipped with the ℓ^1 direct sum $(V \oplus_1 W, \|(v, w)\| = \|v\| + \|w\|)$. This categorical structure intrinsically encodes additivity over local edges. The space of 1-cochains $C^1(P, F) = \bigoplus_{e \in \mathrm{E}_{\mathrm{cov}}} F(a_e)$ equipped with the ℓ^1 norm is the free additive Banach object generated by the edges. By Axiom 1, any diagnostic factorizes through the coboundary map $\delta \colon \Gamma(F) \to C^1(P, F)$.

Theorem 4.3 (ℓ^1 Rigidity)**.** *The ℓ^1 coboundary norm $I(s) = \sum_e \|\delta_e(s)\|$ is the* ***free additive faithful contractive seminorm*** *on $C^1(P, F)$. It is the* ***initial object in the category of additive faithful contractive seminorms on*** *$C^1(P, F)$. Consequently, any diagnostic J_P satisfying Axioms 1–4 factors uniquely as $J_P(s) = \alpha \cdot I(s)$ for some $\alpha > 0$.*

Proof. By defining the metric through the Hahn–Banach scaling of contractive monotonic maps, exact 1-homogeneity natively limits variations to uniform posetic scalars alone. The full categorical argument is given in Appendix C (forthcoming).
Thus ℓ^1 is not merely a convenient choice of diagnostic; it is forced by the structural axioms governing local defect aggregation.

Remark 4.4 (Universal Property). Re-framing the uniqueness theorem as a universal property addresses why ℓ^1 specifically emerges: it is the *unique additive geometry* compatible with the axioms. The ℓ^2 norm fails because $\|(v, w)\|_2 \neq \|v\| + \|w\|$, while ℓ^∞ fails because $\max(\|v\|, \|w\|)$ discards summation over independent local defects. The ℓ^1 norm is structurally forced because it is the universal map out of the free space of local inconsistencies (the Kan extension $\mathrm{Lan}_\pi d(s)$).

5. Theorem III: Application to Density Scaling

5.1. **Setup.** Consider N sites on a finite graph with **finite-dimensional** local Hilbert spaces ($\dim \mathcal{H}_x \leq d < \infty$), Hamiltonian $H = \sum_e H_e$ with $\|H_e\| \leq J$, and product-state projection $\Pi(\rho) = \bigotimes_x \mathrm{Tr}_{\bar{x}}(\rho)$. Since H is bounded, all domain conditions of Theorem 3.1 are automatic.

5.2. **Explicit Rate Bound.**

Lemma 5.1 (Local obstruction rate)**.** *For a single bond $e = (a, b)$ with $[H_e, \rho_a \otimes \rho_b] \neq 0$, define $\gamma_e = \|[H_e, \rho_a \otimes \rho_b]_{\mathrm{conn}}\|_1 / d^2$. Then $\gamma_e > 0$ and $\gamma_e \leq 2J$. For t small:*

$$C_e(t) \geq \gamma_e t - O(J^2 t^2).$$

Proof. On the finite-dimensional space $\mathcal{H}_e$ ($\dim \leq d^2$), the map $t \mapsto \rho_e(t) - \rho_a(t) \otimes \rho_b(t)$ is entire analytic. Its leading Taylor coefficient is $\|[H_e, \rho_a \otimes \rho_b]_{\mathrm{conn}}\|_1$, which is positive by hypothesis. The denominator d^2 converts trace-norm to operator-norm constraints: $\|A\|_1 \leq d^2 \|A\|_{\mathrm{op}}$. The remainder is $O(J^2 t^2)$ since $\|H\| \leq zJ$ and higher-order commutators grow at most geometrically. □

5.3. **The Density Bound.**

Theorem 5.2 (Finite-Time Density Scaling)**.** *Let $\rho(0) = \bigotimes_x \rho_x(0)$ with fraction $\eta > 0$ of bonds satisfying $[H_e, \rho_a \otimes \rho_b] \neq 0$. Define $\gamma_* = \min_{e \in E_{\mathrm{active}}} \gamma_e > 0$ (attained: E_{active} is finite). Then $C(t) \geq c > 0$ for $t \in [t_1, t_2]$, where c, t_1, t_2 depend only on J, z, d—not on N.*

Proof sketch. By Lemma 5.1, each active bond has $C_e(t) \geq \gamma_e t - O(J^2t^2) > 0$ for $t \in (0, \gamma_*/(2zJ)^2)$. Continuity gives a positive interval $[t_1, t_2]$ with $C_e(t) \geq \frac{1}{2}\gamma_* t_1$ for all active e. By Lieb–Robinson [1, 6], with finite propagation velocity $v_{\mathrm{LR}} = O(Jz)$, bulk bonds (graph distance $\geq v_{\mathrm{LR}}t$ from the boundary) have marginals independent of N up to exponentially small corrections. Summing over the macroscopic number of active bonds: $C(t) \geq \eta \cdot \frac{1}{2}\gamma_* t_1 = \Omega(1)$. □

6. Worked Examples

6.1. **Example 1: Two Qubits (Minimal).** Two qubits ($d = 2$) with $H = J \cdot Z \otimes Z$ and initial product state $\rho(0) = |0\rangle\langle 0| \otimes |+\rangle\langle +|$. The commutator:

$$[Z \otimes Z,\ |0\rangle\langle 0| \otimes |+\rangle\langle +|] = |0\rangle\langle 0| \otimes [Z, |+\rangle\langle +|] \neq 0.$$

Exact continuous-time evolution $\rho(t) = e^{-iJtZ\otimes Z}\rho(0)e^{iJtZ\otimes Z}$ yields $C(t) = \sin^2(Jt)$. This satisfies $C(t) > 0$ for $0 < t < \pi/J$—a single-bond illustration of Theorem 5.2. The obstruction's existence ($C > 0$) follows from Theorem 3.1; its measurement by $\|\cdot\|_1$ is forced by Theorem 4.3.

6.2. **Example 2: Transverse-Field Ising Model.** The 1D transverse-field Ising model $H = -J\sum_i Z_iZ_{i+1} - h\sum_i X_i$ on N qubits with $\rho(0) = |+\rangle\langle +|^{\otimes N}$. Since $\langle +|Z|+\rangle = 0$, the first-order contribution vanishes: $C_e(t) = O(J^2t^2)$ with strictly positive leading coefficient. By translational invariance, $C(t) = O(J^2t^2) > 0$ for $t > 0$, with per-bond density insensitive to N for $t \ll N/v_{\mathrm{LR}}$.

6.3. **Example 3: Classical Factorized Dynamics.** Coupled oscillators $\dot{x}_i = -\omega_i^2 x_i + \epsilon\sum_{j\sim i}(x_j - x_i)$ with factorized initial distribution. The coupling generates correlations at rate $\epsilon \cdot \mathrm{Var}(x_j) > 0$, giving $C_{\mathrm{cl}}(t) = O(\epsilon^2t^2)$ with coefficient independent of N.

7. Discussion

7.1. **Correlation Flux Interpretation.** The coboundary $\delta_e(s) = r_{b,a}(s(b)) - s(a)$ quantifies the mismatch between local states across the edge e. In physical settings this mismatch corresponds to the **generation of correlations** between subsystems. Under this interpretation:

- **Theorem 3.1:** Projection creates non-removable correlation sources.
- **Theorem 4.3:** ℓ^1 uniquely measures total flux magnitude as the universal map out of the free defect space.
- **Theorem 5.2:** Flux propagates locally at finite speed.

Relation to entanglement. The obstruction $C_e(t) = \|\rho_{ab} - \rho_a \otimes \rho_b\|_1$ bounds mutual information via Fannes–Audenaert [15]: $I(A{:}B) \leq 2\varepsilon \log d + 2h(\varepsilon)$. Its early-time scaling $O(J^kt^k)$ matches entanglement entropy growth [13, 14], since both are controlled by the same Dyson-series commutators.

7.2. **Domain of Validity.** Theorems 3.1–4.3 apply to Banach presheaves over finite posets. Theorem 5.2 requires finite-dimensional local Hilbert spaces and finite-range non-commuting interactions. The framework does not apply to infinite-dimensional posets, long-range interactions, or commuting Hamiltonians.

7.3. **Relation to Existing Work.** The ℓ^1 coboundary norm is standard in Čech cohomology; the novelty is the uniqueness theorem (Theorem 4.3) showing it is forced by axioms. Our framework generalizes the sheaf-theoretic contextuality of Abramsky–Brandenburger [8] from boolean to Banach presheaves: binary contextuality (sections glue or not) becomes quantitative (ℓ^1 measures how badly gluing fails).

7.4. **Covering Map Transport.** The ℓ^1 norm satisfies $I_{\tilde{P}}(\pi^* s) = k \cdot I_P(s)$ for degree-k covering maps (Theorem A.1) but is **not** preserved under general order-preserving maps (Appendix B).

7.5. **Future Directions.** Two extensions are in preparation:

- **Free seminorm classification:** A comprehensive categorical proof that ℓ^1 is the free additive faithful contractive seminorm.
- **Universal obstruction theory:** Extension to Lindbladian dynamics and scalar propagation.

Appendix A. Covering Map Extensivity

Theorem A.1. *For a covering map $\pi\colon \tilde{P} \to P$ of degree k: $I_{\tilde{P}}(\pi^* s) = k \cdot I_P(s)$.*

Proof. Each edge has exactly k preimages with identical coboundaries. □

Appendix B. Dimensional Collapse

Theorem B.1. *For a non-covering order-preserving map $f\colon P_1 \to P_2$, there exist F, s with $I_{P_1}(f^* s) \neq I_{P_2}(s)$.*

Proof. Take $P_1 = L_x \times L_t$, $P_2 = L_t$, $f(i,j) = j$. Spatial edges are annihilated; temporal edges amplified by L_x. □

References

[1] E. H. Lieb and D. W. Robinson, "The Finite Group Velocity of Quantum Spin Systems," *Commun. Math. Phys.* **28** (1972), 251–257.
[2] M. Troyer and U.-J. Wiese, "Computational Complexity and Fundamental Limitations to Fermionic Quantum Monte Carlo Simulations," *Phys. Rev. Lett.* **94** (2005), 170201.
[3] S. Mac Lane, *Categories for the Working Mathematician*, Springer, 1971.
[4] A. Grothendieck, *Revêtements Étales et Groupe Fondamental* (SGA 1), Lecture Notes in Mathematics **224**, 1971.
[5] J. Klassen and B. M. Terhal, "Two-local qubit Hamiltonians: when are they stoquastic?," *Quantum* **3** (2019), 139.

[6] B. Nachtergaele and R. Sims, "Lieb–Robinson Bounds and the Exponential Clustering Theorem," *Commun. Math. Phys.* **265** (2006), 119–130.
[7] S. Bravyi, M. B. Hastings, and F. Verstraete, "Lieb–Robinson Bounds and the Generation of Correlations and Topological Quantum Order," *Phys. Rev. Lett.* **97** (2006), 050401.
[8] S. Abramsky and A. Brandenburger, "The sheaf-theoretic structure of non-locality and contextuality," *New J. Phys.* **13** (2011), 113036.
[9] V. Pestov, "Dynamics of infinite-dimensional groups and Ramsey-type phenomena," *Publicações Matemáticas do IMPA*, 2006.
[10] A. Isidori, *Nonlinear Control Systems*, 3rd ed., Springer, 1995.
[11] B. Jakubczyk and W. Respondek, "On linearization of control systems," *Bull. Acad. Pol. Sci. Sér. Sci. Math.* **28** (1980), 517–522.
[12] S. Lang, *Fundamentals of Differential Geometry*, Springer, 1999.
[13] P. Calabrese and J. Cardy, "Evolution of entanglement entropy in one-dimensional systems," *J. Stat. Mech.* (2005), P04010.
[14] H. Kim and D. A. Huse, "Ballistic Spreading of Entanglement in a Diffusive Nonintegrable System," *Phys. Rev. Lett.* **111** (2013), 127205.
[15] K. M. R. Audenaert, "A sharp continuity estimate for the von Neumann entropy," *J. Phys. A* **40** (2007), 8127–8136.
[16] M. Golubitsky and V. Guillemin, *Stable Mappings and Their Singularities*, Springer, 1973.